General Science - Class 4

Solved Exercises - Complete Book (Punjab Curriculum and Textbook Board, Lahore)

Complete-Book Coverage

Complete-book coverage: Chapter 1 Characteristics and Life Process of Organisms (book pp 1-19), Chapter 2 Ecosystem (pp 20-31), Chapter 3 Human Health (pp 32-43), Chapter 4 Matter and Its Characteristics (pp 44-55), Chapter 5 Forms of Energy and Energy Transfer (pp 56-66), Chapter 6 Force and Motion (pp 67-80), Chapter 7 Earth and Its Resources (pp 81-92), Chapter 8 Earth's Weather and Climate (pp 93-99), Chapter 9 Solar System and Our Earth (pp 100-111), Chapter 10 Technology in Everyday Life (pp 112-126), and the Glossary (pp 127-128). Every chapter's Tick the correct answer, short answers, Constructed Response Questions, Investigate and Project items are covered.

Chapter 1 - Characteristics and Life Process of Organisms

1. Tick the correct answer.

Note: in the book, question (i) prints the label (c) twice - once for bones and once for wings. The correct answer is wings (the second (c) option); it is shown here with standard (a)-(d) lettering.

i. Which characteristic is common among butterfly, bird and bat?

(a) teeth

(b) hair

(c) bones

(d) wings ✓

ii. Many plants produce fruits:

(a) to protect seeds. ✓

(b) to produce food for the seeds.

(c) to store water for the seed germination.
(d) to stop seeds from dispersal.

iii. Which one of the following is an example of invertebrate?
(a) cat
(b) butterfly
(c) lizard ✓
(d) frog

iv. Which part of a plant is absent in non-flowering plants?
(a) root
(b) seed
(c) fruit
(d) leaf ✓

v. Which one of the following is a non-flowering plant?
(a) apple
(b) rose
(c) mango
(d) pine ✓

vi. Which statement is correct for all the vertebrates?
(a) Have fur
(b) Have more than four legs
(c) Have backbone ✓
(d) Can fly in air

2. Write short answers.

i. Write any four characteristics of living things?
Living things use food for energy, they breathe, they can move, and they reproduce. They also grow and they can sense changes around them.

ii. Write any three differences between plants and animals?
Plants make their own food while animals depend on other living things for food. Plants cannot move from place to place while animals can move. Plants have roots, stems and leaves, while animals have sense organs and organs for movement.

iii. Differentiate between vertebrates and invertebrates.
Vertebrates are animals that have a backbone, such as fish, frogs, birds and humans. Invertebrates are animals that do not have a backbone, such as insects, worms and snails.

iv. Write names of different types of teeth and their functions.

Incisors are the front teeth used for cutting food. Canines are the pointed teeth used for tearing food. Premolars and molars are the back teeth used for grinding and chewing food.

v. What functions do bones and muscles perform together?

Bones and muscles work together to produce movement. Muscles pull the bones at the joints, so the body can move, walk, run and lift things.

vi. Describe the functions of lungs and heart?

The lungs take in oxygen from the air and remove carbon dioxide from the body during breathing. The heart pumps blood to all parts of the body, carrying oxygen and food to the cells.

3. Constructed Response Questions.

(a) What does a tiger do with its canines?

A tiger uses its large, sharp canines to catch its prey and to tear the flesh of the animals it eats.

(b) What does a rat do with its incisors?

A rat uses its large incisors to gnaw and cut hard food and other materials into small pieces.

4. Investigate.

i. How are the invertebrates useful for humans?

Bees pollinate flowers and give honey, earthworms make the soil fertile, and silkworms give silk. Many invertebrates are also food for other animals.

ii. What is the importance of biodiversity?

Biodiversity keeps the ecosystem balanced. Different kinds of plants and animals provide food, clean air, medicines and raw materials, and every kind has a role in nature.

iii. What is the importance of a flower?

A flower is the reproductive part of a plant. It produces seeds and fruits, attracts insects for pollination, and adds beauty and fragrance to nature.

5. Project: Simulating the heart's pumping of blood.

Project activity: stretch a cut balloon over the mouth of a jar half-filled with coloured water, fit two straws through holes in the balloon, seal one straw tip, and press the balloon again and again. The coloured water comes out of the open straw, just as the heart pumps blood through the body.

Chapter 2 - Ecosystem

1. Tick the correct answer.

i. What is an ecosystem?

(a) System of non-living things is an environment.

(b) Area having a group of living and dead things.

(c) System of living things in an environment.

(d) The collection of abiotic and biotic components in an area. ✓

ii. All the biotic components are:

(a) animals

(b) producers

(c) living things ✓

(d) non-living things

iii. Food chain:

(a) begins with producer ✓

(b) begins with consumer

(c) begins with decomposer

(d) ends with producer

iv. For the conservation of ecosystem:

(a) forests are being cut

(b) roads are being built

(c) tree plantation is being done ✓

(d) factories are being installed

v. If insecticides are used for controlling the population of insects, the population of birds will:

(a) increase

(b) decrease

(c) decrease first then will increase later ✓

(d) increase first, then will decrease later

2. Write short answers.

i. Define environment.

The environment is everything that surrounds a living thing, including all living and non-living things around it, such as air, water, soil, plants and animals.

ii. Write the names of three biotic components of an ecosystem.

The three biotic components of an ecosystem are producers (green plants),

consumers (animals) and decomposers (bacteria and fungi).

iii. Write the names of three abiotic components of an ecosystem.

Three abiotic components of an ecosystem are air, water and soil. Light and temperature are also abiotic components.

iv. Draw a simple food chain.

A simple food chain is: grass (producer) -> grasshopper (consumer) -> frog (consumer) -> snake (consumer) -> hawk (consumer). Energy passes from one living thing to the next in this order.

v. If the food resources increase, what will be the effect on the population of predators?

If food resources increase, the population of prey animals grows, so predators get more food and their population also increases.

vi. Write two human activities which are affecting the ecosystem.

Two human activities affecting the ecosystem are cutting down forests (deforestation) and polluting air, water and land with smoke, chemicals and waste.

3. Constructed Response Questions.

What is the relationship between biodiversity and competition among living things in a balanced ecosystem?

In a balanced ecosystem, many kinds of living things share food, water and space, so they compete with one another. This competition keeps every population in check, and biodiversity keeps the ecosystem stable and healthy.

4. Investigate: Ostriches.

(a) Why does the female ostrich warm the eggs at daytime?

The female ostrich is grey-brown in colour, so during the day she blends with the sandy ground and enemies cannot easily see her while she sits on the eggs.

(b) Why does the male ostrich warm the eggs at night?

The male ostrich is black in colour, so at night he blends with the darkness and enemies cannot easily see him while he sits on the eggs.

5. Project: Make an ecosystem model.

Project activity: cut two bottles, pour water in one to show an aquatic ecosystem and put soil and a small plant in the other to show a land ecosystem. Connect the two with a thick thread so the plant receives water

and minerals, then observe daily and write a report.

Chapter 3 - Human Health

1. Tick the correct answer.

i. If our body temperature increases from 98.6 degrees F to 101 degrees F then its cause is:

(a) hot weather
(b) fever ✓
(c) sitting in the sun
(d) sitting near fire

ii. Which of the following has the highest amount of Calcium?

(a) meat
(b) rice
(c) milk ✓
(d) pulse

iii. If you have flu, what you will do to keep others safe from infection?

(a) will exercise
(b) will sleep for more time
(c) will sit in the sun
(d) will wear mask ✓

iv. Which one of the following causes polio?

(a) bacteria
(b) virus ✓
(c) house fly
(d) mosquito

v. Mainly protection against contagious diseases is done by:

(a) wearing mask, washing hand, vaccination ✓
(b) wearing mask, washing hand, sunbathing
(c) washing hands, sunbathing, sleeping more
(d) vaccination, washing hand, staying indoors

2. Write short answers.

i. Write any three reasons of illness.

Three reasons of illness are germs entering the body, eating an unbalanced diet, and living in dirty surroundings or drinking dirty water.

ii. Differentiate between contagious and non-contagious diseases.

Contagious diseases spread from one person to another through germs,

such as flu and measles. Non-contagious diseases do not spread from person to person, such as diabetes and heart disease.

iii. How does coughing benefit the body?

Coughing clears dust, germs and mucus out of the air passages, so the lungs stay clean and breathing becomes easier.

iv. How does vaccination protect us from diseases?

Vaccination trains the body's defence system to recognise and fight particular germs, so when the real germs enter the body, it can destroy them before we fall ill.

v. What is meant by a balanced diet?

A balanced diet is a diet in which all the components of food - carbohydrates, proteins, fats, vitamins, minerals and water - are present in proper amounts.

3. Constructed Response Questions: Drinking water filtration plant.

i. What is the function of a filter?

A filter removes dirt and solid impurities from water, so the water becomes clean and clear for drinking.

ii. Does filtration also remove the germs?

Filtration removes dirt but it does not remove all the germs from water, so filtered water still needs further treatment to be fully safe.

iii. How can germs be killed in water?

Germs in water can be killed by boiling the water or by adding chlorine to it.

iv. How can water be made safe for drinking?

Water can be made safe for drinking by filtering it to remove dirt, then boiling it or chlorinating it to kill germs, and keeping it covered in a clean container.

4. Investigate.

Activity: interview a doctor or a health worker and ask about the principles of living a healthy life, then write their responses in the table given in the book. This is the student's own interview work.

5. Project.

Activity: observe the kitchen of your home, find out the factors which may cause diseases, and write each factor with its way of prevention in the table given in the book.

Chapter 4 - Matter and Its Characteristics

1. Tick the correct answer.

i. Which one of these has the most volume?

(a) book

(b) pencil

(c) scale

(d) cricket bat ✓

ii. Which one of these groups is the correct example of the three states of matter?

(a) snow, rain, cloud

(b) dew, rain, water vapours

(c) snow, cloud, steam ✓

(d) rain, water vapours, cloud

iii. Put two spoons made of steel and wood in a cup of hot water. After few minutes the steel spoon becomes hot whereas the wooden spoon does not become hot. What does it show?

(a) The steel became hot soon in the presence of wood.

(b) Metal is a better conductor of heat than wood. ✓

(c) Wood is a better conductor of heat than metal.

(d) Metal heats the water quickly than wood.

iv. A piece of ice has been put into a glass of water. Which picture is showing the correct position of the ice?

(a) Picture A: the ice has sunk to the bottom of the glass

(b) Picture B: the ice is in the middle of the water

(c) Picture C: the ice is floating at the surface of the water ✓

(d) Picture D: the ice is floating above the rim of the glass

v. Water, ice and steam all have different temperatures. What is the order from coldest to hottest?

(a) steam, ice, water

(b) ice, steam, water

(c) steam, water, ice

(d) ice, water, steam ✓

vi. Study the table. Matter No. 1 transmits heat quickly, is a solid and does not dissolve in water. Matter No. 2 transmits heat slowly, is a solid and dissolves in water. According to the table which statement is correct about matter No. 1 and matter No. 2?

(a) Matter No. 1 is glass and matter No. 2 is soil.
(b) Matter No. 1 is copper and matter No. 2 is wood.
(c) Matter No. 1 is iron and matter No. 2 is sugar.
(d) Matter No. 1 is cork and matter No. 2 is gold.

2. Write short answers.

i. Define matter and write the name of its three states.

Matter is anything that has mass and occupies space. Its three states are solid, liquid and gas.

ii. Write the differences between solids and liquids.

Solids have a fixed shape and a fixed volume, while liquids have a fixed volume but no fixed shape - they take the shape of their container. The particles of solids are closely packed, while the particles of liquids are less tightly packed and can move.

iii. Which state of matter has lowest density?

Gas has the lowest density, because its particles are far apart and it has very little mass in a given volume.

iv. Write the arrangement of particles in solid.

In a solid the particles are very closely packed in fixed positions. They cannot move from place to place; they only vibrate in their positions.

v. Why are the cooking utensils made up of metals?

Cooking utensils are made of metals because metals are good conductors of heat, so food cooks quickly, and metals are strong and do not break easily.

3. Constructed Response Question.

i. Why do the electricians wear rubber gloves while repairing electric switchboard?

Electricians wear rubber gloves because rubber is an insulator of electricity. The gloves stop electric current from passing into their bodies, so they stay safe from electric shock.

4. Investigate.

i. Why is metal used in making the bell?

Metal is used in making a bell because when metal is struck it vibrates and produces a clear, loud sound that carries far.

ii. Why are metals preferred for making ornaments?

Metals are preferred for ornaments because they are shiny, they can be shaped easily, and they do not rust or spoil quickly.

5. Project.

Activity: collect five things made of metals, observe them, and write their metal, appearance, texture and colour in the table given in the book.

Chapter 5 - Forms of Energy and Energy Transfer

1. Tick the correct answer.

i. Which of the following is NOT a form of energy?

(a) Light

(b) Sound

(c) Glass ✓

(d) Heat

ii. We hear echo when sound:

(a) reaches us direct from the source.

(b) comes after bouncing back from a wall at a certain distance. ✓

(c) comes from a loudspeaker.

(d) is very loud.

iii. How many colours are there in sunlight?

(a) 1

(b) 3

(c) 5

(d) 7 ✓

iv. If you take your hand first nearer to a lit lamp and then away from it, the shadow of your hand:

(a) will not form.

(b) will be smaller. ✓

(c) will be larger.

(d) will be of the same size.

v. Electrical energy can be transformed into:

(a) heat. ✓

(b) light.

(c) sound.

(d) Plastic

Note on question (v): heat, light and sound are all correct transformations according to the lesson (a heater changes electrical energy into heat, a bulb into light, and a loudspeaker into sound). Only Plastic is not a form of energy. The tick is placed on heat, with this explanation.

2. Write short answers.

i. Can light, sound and heat travel through vacuum?

Light can travel through vacuum, and heat can travel through vacuum by radiation. Sound cannot travel through vacuum because it needs a medium, such as air, to travel through.

ii. If sound cannot travel through vacuum, how do the astronauts talk with each other?

Astronauts talk with each other using radio waves, because radio waves are a form of energy that can travel through vacuum, unlike sound.

iii. How do we sense the buzzing of a mosquito?

The mosquito's wings vibrate and produce sound waves that travel through the air to our ears, and our ears sense these vibrations as a buzzing sound.

iv. Into which two forms of energy is the electrical energy transformed in a television set?

In a television set, electrical energy is transformed into light energy (the picture) and sound energy.

v. When can we see rainbow? How is it formed?

We can see a rainbow when it rains and the Sun shines at the same time. Sunlight splits into seven colours as it passes through the raindrops, and this band of colours is the rainbow.

3. Constructed Response Questions.

i. If we bring a thermometer near a lighted bulb, will the thermometer show any change in temperature? If yes, would the temperature rise or fall? Explain.

Yes, the thermometer will show a change - the temperature will rise. A lighted bulb gives out heat as well as light, so the air near the bulb becomes warmer and the thermometer reading goes up.

ii. You are holding a glass of cold water in a room. Body temperature 37 degrees C, water temperature 05 degrees C, room temperature 30 degrees C. Draw arrows in the picture to show the directions of the heat flow.

Heat always flows from a hotter thing to a colder thing. So heat flows from your hand (37 degrees C) into the cold water (05 degrees C), and heat also flows from the room air (30 degrees C) into the cold water. Both arrows point into the glass of water.

iii. Sidra has a reflector in her torchlight. Ali's torchlight has no reflector. Which torchlight will shine more light on a wall five metres away? Tick the box for your answer.

Sidra's torch will shine more light on the wall. The reflector gathers the light from the bulb and sends it forward in one direction, so more light reaches the wall.

4. Investigate: Complete the electric circuit.

Which thing or things should we connect at the ends of the wire so that the electric circuit becomes complete and the bulb lights up?

The nail and the key should be connected at the ends of the wire. Both are made of metal, and metals are conductors of electricity, so the circuit becomes complete and the bulb lights up. The cotton bud, plastic ball pen and ice-cream stick are insulators and will not complete the circuit.

5. Project: Let us make a telephone.

Project activity: make holes in two paper cups, join them with a string or thin wire about seven metres long, pull the string tight, and speak into one cup while a friend listens at the other. Sound vibrations travel along the tight string.

Chapter 6 - Force and Motion

1. Tick the correct answer.

i. What is force?

(a) Push

(b) Pull

(c) Lift

(d) Push and pull ✓

ii. When a body changes its position, it is called:

(a) friction

(b) force

(c) motion ✓

(d) gravity

iii. All bodies are attracted towards the Earth due to:

(a) force

(b) friction

(c) gravity ✓

(d) motion

iv. If we apply the same force on a toy car, it will travel longer on:

(a) rocky soil

(b) marble floor

(c) floor of bricks ✓

(d) ground

v. Which machine consists of a grooved wheel?
(a) Lever
(b) Pulley
(c) Inclined plane
(d) Gear

2. Write short answers.

i. How force and motion are related? Explain.
Force and motion are closely related: applying a force can start motion in a body at rest, stop a moving body, change its speed, or change its direction. Without a force, a body does not change its state of motion.

ii. What is gravity? On which objects does it act?
Gravity is the force with which the Earth pulls everything towards itself. It acts on all objects, whether they are near the Earth or far from it.

iii. Define friction. In which direction does it act?
Friction is the force that opposes motion when two surfaces rub against each other. It always acts in the opposite direction to the motion.

iv. What is a machine? How does it work for us?
A machine is a device that makes our work easier. It works for us by changing the size or the direction of the force we apply, so we can lift heavy loads or do work with less effort.

v. Why cannot we walk on ice easily?
We cannot walk on ice easily because ice is very smooth, so there is very little friction between our shoes and the ice, and our feet slip.

3. Constructed Response Questions.

i. Look at these shoes. Which one is suitable for walking on a rocky surface? Which one would you use in the playground? Which one is for using on an icy surface? Explain the reasons in your answers.
The strong, hard-soled shoe with a deep tread is suitable for a rocky surface, because its grip protects the feet and stops slipping on rough stones. The light, comfortable sports shoe is for the playground, because it is soft and easy to run and play in. The shoe with a rough or spiked sole is for an icy surface, because it increases friction and stops the feet from slipping on ice.

ii. This patient is sitting in a wheelchair. Which machine will be useful to take him on the stage? Why?

An inclined plane (a ramp) will be useful to take the patient on the stage. A ramp lets the wheelchair be pushed up gradually with much less force than lifting it straight up.

4. Investigate.

(a) What force do you use (push or pull) while writing?

While writing, we use a push force - we push the pencil forward across the paper to form the letters.

(b) What is the role of friction in writing on the paper with a pencil?

Friction between the pencil tip and the paper rubs off tiny particles of graphite that leave the mark on the paper, and friction also gives the grip that lets us control the pencil.

5. Project: Making a pulley for hoisting a flag.

Project activity: make a flagpole in the school playground with a bamboo pole, an iron hanger, a small wooden reel, a rope, the flag of Pakistan, a plumb line and some bricks, then hoist the national flag with this pulley.

Chapter 7 - Earth and Its Resources

1. Tick the correct answer.

i. Glass is made from:

(a) soil.
(b) sand.
(c) salt. ✓
(d) rubber.

ii. About what percentage of the Earth's surface is land?

(a) 1 percent
(b) 21 percent
(c) 29 percent ✓
(d) 30 percent

iii. The solid non-living chemicals present in the Earth are:

(a) soil.
(b) minerals.
(c) stones. ✓
(d) fossils.

iv. To make things reusable is called:

(a) recycling. ✓
(b) conservation of resources.

(c) care.
(d) cycling.

v. An example of non-renewable resources is:
(a) oil.
(b) soil.
(c) air.
(d) solar energy.

2. Write short answers.

i. What are fossils? About which organisms do they provide information?
Fossils are the remains or imprints of dead plants and animals preserved in rocks. They provide information about the organisms that lived on the Earth millions of years ago, including organisms that are now extinct.

ii. Why are forests called renewable resources?
Forests are called renewable resources because trees can be grown again. If trees are cut down and new trees are planted, the forest can be replaced in a reasonable time.

iii. How does water reach oceans and lakes?
Water reaches oceans and lakes through rivers and streams that flow downhill into them, and also directly through rain and the melting of snow and glaciers.

iv. Describe any two advantages and disadvantages of deforestation.
Two advantages of deforestation are that land becomes available for farming and houses, and wood is obtained for fuel and furniture. Two disadvantages are that animals lose their homes, and the soil is washed away and rainfall decreases.

v. Differentiate between renewable and non-renewable resources.
Renewable resources can be replaced naturally in a short time, such as forests, water and solar energy. Non-renewable resources take millions of years to form and cannot be replaced once used up, such as coal, oil and natural gas.

3. Constructed Response Questions.

i. We found the skulls of various animals during an excavation. How do we know about their feed?
We know about their feed from the shape of their teeth. Sharp canines show the animal ate meat, while large flat molars and incisors show the animal ate plants.

ii. Jellyfish is a soft invertebrate. Can we obtain a fossil of jellyfish?

No, we normally cannot obtain a fossil of a jellyfish. Its body is soft and has no bones or shell, so it decays completely after death instead of being preserved in rocks.

4. Investigate.

i. Is there anything that is not made from natural resources?

Almost everything we use is made from natural resources. Even things made in factories come from materials taken from nature, such as metals from rocks, wood from trees, and plastic from oil.

ii. Water is a renewable resource. There should be plenty of water for every human being on the Earth but why is it not so?

Water is not equally available everywhere: much of it is salty sea water, many places get little rain, and water is wasted and polluted, so many people do not get enough clean drinking water.

iii. Is it possible that we have more resources than those of the past?

No, we cannot have more resources than in the past. Non-renewable resources are being used up and cannot be replaced, so we must use all resources carefully and save them.

5. Project.

Project activity: make a model of the flow of water from a glacier to the sea using suitable materials, showing how melting ice feeds rivers that carry water to the sea.

Chapter 8 - Earth's Weather and Climate

2. Write short answers.

i. Differentiate between weather and climate.

Weather is the condition of the atmosphere of a place at a particular time, such as a hot, rainy or cloudy day. Climate is the general weather condition of a region over a long period of time.

ii. What is the relation between climate and the height from sea level?

The higher a place is above sea level, the colder its climate becomes. That is why mountain areas are colder than places near the sea coast.

iii. Why does climate change with distance from the Equator?

Places near the Equator receive direct rays of the Sun and stay hot, while places far from the Equator receive slanting rays spread over a larger area, so they are colder.

iv. Why is the climate in the Polar Zone extremely cold?

The Polar Zone is extremely cold because the Sun's rays fall very slantingly there and much of the heat is reflected back by ice and snow, so very little warmth remains.

v. Based on climate, in which zone is Pakistan located?

Based on climate, Pakistan is located in the North Temperate Zone, which has a moderate climate with hot summers and cool winters, while its northern mountains are cold.

3. Constructed Response Questions.

i. Identify the different zones on the figure of the Earth, on the basis of climate.

On the basis of climate, the Earth is divided into the Torrid Zone around the Equator, the Temperate Zones (North and South) in the middle, and the Frigid or Polar Zones around the North and South Poles.

ii. What is the climate of these zones? Describe the reasons.

The Torrid Zone is hot all year because the Sun shines directly overhead.
The Temperate Zones are moderate because the Sun's rays fall slantingly.
The Frigid Zones are extremely cold because the rays fall very slantingly and ice reflects most of the heat.

iii. In winter, the temperature of Skardu, a city of Gilgit-Baltistan is below zero degree centigrade while the temperature of Karachi is mild. Why is it so?

Skardu is a high mountain city far from the sea, so its winter temperature falls below zero. Karachi is at sea level near the coast, and the sea keeps its temperature mild even in winter.

4. Investigate.

How many weather conditions are there on the Moon? Does the Moon also have storms, rains, and snowfall like the Earth? What kind of a climate is there on the Moon?

There are no weather conditions on the Moon, because the Moon has almost no atmosphere. So there are no storms, rains or snowfall on the Moon. Its climate is extremely harsh - very hot in sunlight and extremely cold in darkness.

5. Project.

Project activity (i): construct a simple wind vane using cardboard or old carton, scissors, straw, board pins and coloured chart paper, as shown in

the pictures in the book.

Project activity (ii): make a simple rain gauge from a transparent plastic bottle, scissors, adhesive tape and a paper strip marked in millimetre scale, and use it to compare the amount of rain that fell in three days during a rainy season.

Chapter 9 - Solar System and Our Earth

1. Tick the correct answer.

i. The gravity of which body keeps the planets and other celestial bodies together in the Solar System?

(a) Jupiter

(b) Earth

(c) Sun ✓

(d) Moon

ii. The Earth completes its rotation around the Sun in 365 days. This period is called:

(a) a solar year ✓

(b) a solar month

(c) lunar month.

(d) lunar year.

Note on question (ii): the book prints the word 'rotation' here, where revolution (movement around the Sun) is meant. The period of 365 days is a solar year.

iii. Due to the annual revolution of Earth and tilt of its axis:

(a) day and night are formed.

(b) seasons change. ✓

(c) a lunar month

(d) a lunar year

iv. On the globe, in which part of the Earth is Pakistan situated?

(a) Northern hemisphere ✓

(b) Southern hemisphere

(c) Equator

(d) Half in the northern hemisphere and half in the southern hemisphere

v. Which planet of the Solar System does not have any moon?

(a) Jupiter

(b) Venus ✓

(c) Mars

(d) Saturn

2. Write short answers.

i. We can only see one side of the Moon. Explain.

We can only see one side of the Moon because the Moon takes the same time to spin on its own axis as it takes to revolve around the Earth, so the same side always faces the Earth.

ii. If there were no tilt in the Earth's axis, how would it have affected the seasons?

If there were no tilt in the Earth's axis, there would be no change of seasons. Every place would have the same weather all year round.

iii. When it is winter in the Northern Hemisphere then what will be the season in Southern Hemisphere?

When it is winter in the Northern Hemisphere, it is summer in the Southern Hemisphere, because the two hemispheres receive the Sun's rays oppositely.

iv. Which planet is closest to the Sun?

Mercury is the planet closest to the Sun.

v. Why is the Solar Eclipse usually partial?

A solar eclipse is usually partial because the Moon's dark shadow is very small and rarely falls fully on one place, and the alignment of the Sun, Moon and Earth is seldom perfect, so most places see only a part of the Sun covered.

3. Constructed Response Questions.

i. Which phenomena are shown in the following figure?

The figures show a solar eclipse and a lunar eclipse. In a solar eclipse the Moon comes between the Sun and the Earth, and in a lunar eclipse the Earth comes between the Sun and the Moon.

ii. In the following picture, each arrowhead shows the axial or orbital movement of the Earth and the Moon. Write the correct type of movement in the box given near each arrowhead.

The Earth's spin on its own axis is its axial movement (rotation), and its movement around the Sun is its orbital movement (revolution). Likewise, the Moon's spin on its axis is axial movement, and its movement around the Earth is orbital movement.

iii. In how many days, the Moon completes its one revolution around the Earth?

The Moon completes its one revolution around the Earth in about 27 days.

iv. In how many days, the Earth completes its one revolution around the Sun?

The Earth completes its one revolution around the Sun in 365 days.

Identify the season in the southern hemisphere as per given figure. Write the correct season in the boxes given near the dates.

In the southern hemisphere: on 21 March it is autumn, on 22 June it is winter, on 23 September it is spring, and on 22 December it is summer - the opposite of the northern hemisphere.

4. Investigate.

If the Sun stops providing light and heat, what will be its effects on the Earth? Would life be possible on the Earth?

If the Sun stopped providing light and heat, the Earth would become dark and extremely cold, plants would die because they could not make food, and animals and humans would soon die too. Life would not be possible on the Earth without the Sun.

5. Project.

Project activity: make two groups in the class. One group will make the model of a lunar eclipse and the other group will make the model of a solar eclipse.

Chapter 10 - Technology in Everyday Life

1. Tick the correct answer.

i. Clinical thermometer is used to:

(a) check sugar

(b) measure inflamation in the body

(c) check fever ✓

(d) measure blood pressure

ii. The number of corners of a prism are:

(a) 3

(b) 4

(c) 5

(d) 6 ✓

iii. The surfaces of a cube are:

(a) 3

(b) 4

(c) 6 ✓

(d) 8

iv. Taking photographs of oneself is known as:
(a) portrait
(b) selfie
(c) landscape
(d) oneself

v. Which item is used to reduce inflammation as a first aid?
(a) medical tape
(b) tweezer
(c) thermometer
(d) instant cold pack

vi. The blood pressure 160/100 is:
(a) low blood pressure
(b) high blood pressure
(c) normal blood pressure
(d) not possible

2. Write short answer.

i. What is the difference between a cone and a prism?
A cone has one circular base and one pointed top, while a prism has two identical flat ends (for example two triangles) joined by rectangular faces. A cone has a curved surface, while a prism has only flat surfaces.

ii. Can an envelope be made from a square-shaped paper? Explain.
Yes, an envelope can be made from a square-shaped paper. By folding the four corners of the square towards the centre and pasting three of them, a pocket is formed that works as an envelope.

iii. Why a line is drawn on a paper before cutting it with a scissors?
A line is drawn on the paper before cutting so that the scissors follow the line and the paper is cut straight and exactly to the required size and shape.

iv. Why a clinical thermometer is shaken for a few times before use?
A clinical thermometer is shaken a few times before use so that the mercury thread falls back below 35 degrees C. This resets the thermometer, so it can correctly measure the patient's body temperature.

v. Why any soil cannot be used to make a model?
Any soil cannot be used to make a model because ordinary soil is loose and crumbly and does not hold its shape. A model needs a soft, sticky material like clay or playdough that keeps the shape given to it.

3. Constructed Response Questions.

i. A circular shape is cut as shown in the following figure. What is the value in fraction of the total in each case? Write below each shape.

The blue circle is cut into 2 equal parts, so each part is 1/2 of the total. The red circle is cut into 3 equal parts, so each part is 1/3. The green circle is cut into 4 equal parts, so each part is 1/4. The yellow circle is cut into 5 equal parts, so each part is 1/5. The purple circle is cut into 6 equal parts, so each part is 1/6.

ii. Identify the various items used in everyday life which look like a circle, a cube, a cylinder, a cone and a prism. Give two examples of each shape.

Circle: a coin and a plate. Cube: a dice and a box. Cylinder: a glass and a tin. Cone: an ice-cream cone and a funnel. Prism: a tent and a Toblerone chocolate box.

iii. Write below each shape, number of its corners, edges and surfaces.

Sphere: 0 corners, 0 edges, 1 surface. Cube: 8 corners, 12 edges, 6 surfaces. Triangular pyramid: 4 corners, 6 edges, 4 surfaces. Cone: 1 corner, 1 edge, 2 surfaces. Triangular prism: 6 corners, 9 edges, 5 surfaces.

4. Investigate.

Why mobile phone technology is progressing rapidly? How will be the mobile phone of the future?

Mobile phone technology is progressing rapidly because people want faster communication, internet, cameras and many uses in one small device, and better chips and networks keep being developed. The mobile phone of the future will be even smarter, thinner and faster, with foldable screens and many more uses in daily life.

5. Project: Making a Nest.

Project activity: take an empty pack of juice or milk, cut a face large enough for a small bird to enter and exit easily, pad the interior with cotton or cloth, drill a hole and insert a wire through the upper edge, cover the pack with colour ribbons and packing paper, then hang the nest near your home or school and observe whether any bird uses it.

Glossary (book pp 127-128) - Reference

The book's glossary is included here as a reference section; it is not an exercise to solve.

Abiotic components

The non-living components of an ecosystem (air, water, light, soil).

Anemometer
The instrument to measure the direction and speed of air (wind).

Axial motion
The movement of the Earth around its own axis.

Balanced diet
The diet in which all components of food are present in proper amounts.

Barometer
The instrument to measure the pressure of air.

Biodiversity
Various kinds of organisms present in a specific region.

Biotic components
The living organisms in an ecosystem (producers, consumers, decomposers).

Climate
The general weather conditions of a region.

Consumers
The organisms which get food from other organisms (for example, animals).

Global warming
The increase in the average temperature of Earth, due to pollution.

Heart
The organ responsible for the circulation of blood in the body.

Hydroelectricity
The electricity produced by the movement of water.

Invertebrates
The animals without a backbone.

Joint
The point where two or more bones join.

Lever
A simple machine used to lift heavy objects.

Light
A type of energy that helps us to see the things of the surroundings.

Liquid
The state of matter that has a specific volume but has no specific shape.

Lungs
The pair of organs responsible for the exchange of gases between blood and air.

Mass
The amount of matter in a body.

Motion
The change in the position of a body.

Notes on the answers (flags)

1. Chapter 1, MCQ (i): the book prints the label (c) twice - once for bones and once for wings. The correct answer is wings (the second (c) option); it is shown here with standard (a)-(d) lettering and the green tick on wings.
2. Chapter 5, MCQ (v): heat, light and sound are all correct transformations according to the lesson (heater changes electrical energy into heat, bulb into light, loudspeaker into sound); only Plastic is not a form of energy. The tick is placed on heat, with this explanation given next to the question.
3. Chapter 9, MCQ (ii): the book prints the word 'rotation' where revolution (movement around the Sun) is meant. The period of 365 days is a solar year.
4. Chapter 4, MCQ (iv): the options in the book are four pictures of a glass of water. They are reproduced here as descriptions (A sunk at the bottom, B in the middle, C floating at the surface, D above the rim); the correct position is C, because ice floats at the surface.
5. Chapter 6, Constructed Response (i), and Chapter 10, Constructed Response (i) and (iii): these questions are figure-based in the book. The answers describe the printed figures (shoes; cut circles; sphere, cube, triangular pyramid, cone, triangular prism).
6. Investigate and Project items that are the student's own work (interviews, observations, models) are summarised as activity notes, not solved answers.
7. The book's glossary (book pp 127-128) is included as a reference section. After the glossary the book has one blank page, an approval page and the back cover, with no further exercises.